

HUMAIN, Belgium

WMO: 064720

Lat: 50.1936N

Lon: 5.2556E

Elev: 296

StdP: 97.82

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.9	(c) -5.6	(d) -11.7	(e) 1.4	(f) -5.5	(g) -8.1	(h) 2.0	(i) -3.9	(j) 12.9	(k) 7.3	(l) 11.7	(m) 7.5	(n) 3.3	(o) 50	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.5	(c) 29.4	(d) 19.9	(e) 26.9	(f) 19.0	(g) 24.9	(h) 18.1	(i) 21.1	(j) 26.9	(k) 20.0	(l) 24.8	(m) 18.9	(n) 23.5	(o) 3.3	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.6	23.2	18.3	13.6	22.1	17.2	12.8	20.8	62.6	27.0	58.4	24.6	54.9	23.7	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.1	8.7	7.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-10.3	23.1	2.5	1.5	-12.1	24.1	-13.5	25.0	-14.9	25.8	-16.7	26.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	2.0	2.4	6.2	9.3	12.2	16.1	18.2	17.2	14.3	10.9	6.2	3.2
	DBStd	6.72	3.89	4.51	4.06	3.95	3.98	3.44	3.40	3.09	3.07	3.84	3.46	3.72
	HDD10.0	1038	248	214	127	59	21	0	0	0	2	34	121	213
	HDD18.3	3176	506	447	376	270	196	86	44	59	128	230	364	471
	CDD10.0	998	0	1	9	39	88	183	253	223	129	63	7	1
	CDD18.3	92	0	0	0	0	4	19	39	23	6	1	0	0
	CDH23.3	820	0	0	0	3	37	174	348	215	42	1	0	0
	CDH26.7	227	0	0	0	0	4	42	107	69	6	0	0	0
(14) Wind	WSAvg	3.7	4.4	4.1	4.2	3.6	3.1	3.3	3.0	3.1	3.1	3.7	3.9	4.4
(15) Precipitation	PrecAvg	1040	107	87	78	62	74	79	88	97	77	84	90	117
	PrecMax	1290	269	209	169	161	171	161	211	241	166	196	151	277
	PrecMin	763	12	13	11	5	30	20	42	23	13	36	5	18
	PrecStd	138	51	43	42	37	35	37	37	56	45	32	37	55
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	15.7	19.2	23.4	26.9	30.4	33.1	31.7	27.2	22.8	15.7	12.1
		MCWB	10.6	7.7	12.4	14.7	18.6	20.0	20.1	20.6	19.5	16.6	13.1	10.2
	2%	DB	10.1	10.8	17.0	21.4	24.0	27.4	29.6	28.5	24.4	20.4	13.9	10.8
		MCWB	9.4	7.5	10.8	13.9	16.5	19.9	20.2	19.4	18.1	15.9	12.5	9.6
	5%	DB	8.5	9.0	14.7	18.9	21.7	25.2	27.0	25.5	22.0	18.1	12.7	9.4
		MCWB	7.9	7.5	10.0	13.1	15.7	18.1	18.8	18.4	16.9	15.0	11.6	8.7
	10%	DB	7.3	7.9	12.3	16.3	19.4	23.0	24.8	23.0	19.8	16.3	11.2	8.0
		MCWB	6.6	6.8	9.0	11.7	14.7	17.3	18.3	17.8	15.9	13.9	10.2	7.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	9.6	12.7	15.9	20.1	22.4	22.2	22.5	20.6	17.8	13.8	11.2
		MCDB	11.2	11.7	17.6	21.6	25.4	27.7	28.3	29.0	24.9	21.1	14.8	11.4
	2%	WB	9.5	8.6	11.7	14.6	17.9	20.7	21.0	20.5	19.0	16.5	12.7	9.7
		MCDB	9.9	9.9	15.6	19.9	22.1	25.8	27.3	25.5	23.3	19.1	13.7	10.4
	5%	WB	8.0	7.6	10.4	13.3	16.5	19.0	19.9	19.1	17.6	15.3	11.7	8.7
		MCDB	8.3	8.9	13.7	17.8	20.1	23.3	25.2	23.3	21.1	17.3	12.6	9.4
	10%	WB	6.7	6.7	9.4	12.1	14.9	17.8	18.9	18.1	16.4	14.4	10.3	7.5
		MCDB	7.2	8.0	12.0	16.0	18.6	22.1	23.9	22.3	19.3	16.1	11.1	8.0
(35) Mean Daily Temperature Range	5% DB	MDBR	3.9	5.2	7.7	9.3	8.8	9.2	9.5	9.0	8.6	6.7	4.6	4.2
		MCDBR	4.8	7.2	12.4	12.7	12.3	12.9	13.0	13.3	11.7	9.3	6.1	5.5
	5% WB	MCWBR	4.5	5.2	7.4	7.3	6.7	6.5	5.8	6.1	6.5	5.6	4.9	4.8
		MCDWR	4.6	6.3	10.2	11.4	10.6	11.2	11.1	10.6	10.3	8.0	5.6	5.1
(40) Clear-Sky Solar Irradiance	taub	0.306	0.330	0.362	0.393	0.403	0.403	0.402	0.396	0.366	0.349	0.323	0.304	0.304
		taud	2.440	2.394	2.321	2.256	2.264	2.284	2.280	2.324	2.388	2.436	2.443	2.414
	Ebn at Noon	738	806	838	848	855	856	849	836	825	772	706	678	678
		Edh at Noon	61	82	104	124	128	127	125	115	97	78	61	55
(44) All-Sky Solar Radiation	RadAvg	0.79	1.48	2.65	4.17	4.94	5.42	5.16	4.40	3.34	1.96	0.93	0.63	0.63
	RadStd	0.12	0.28	0.37	0.50	0.52	0.50	0.58	0.48	0.35	0.17	0.11	0.10	0.10

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page